



smartcare

CARETAKER MANAGEMENT SYSTEM

PROJECT PROPOSAL

SCS 2301-Group Project I

Group 11

Details of Project Supervisor, Co-supervisor, Advisors and Clients

Proposed project supervisor(Academic Staff of UCSC)

Name of the supervisor : Ms.S Hamsavasini

Signature of the supervisor:


.....

Date: 27.06.2025

Proposed Project Co-Supervisor (Assigned by Course Coordinator)

Name of the co-supervisor: Ms.P.H.D.G.N.De Silva

Signature of the co-supervisor:

.....

Date:

The client of the Project

Name of the client:Ms. S Hamsavasini

Address of the client:

Contact person at client:

The contact number of the contact person:

E-mail address of the contact person:

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1. Introduction

1.1. Problem Definition

In today's world, many families need help caring for elderly people, children, or patients at home. Finding the right caretaker who is trusted, skilled, and available is not easy. At the same time, caretaker companies also face challenges such as managing staff, handling client requests, tracking attendance, and keeping up with daily schedules.

Many existing systems do not support flexible planning, performance tracking, or clear communication between caretakers and clients. As a result, clients often feel confused about who is coming, caretakers may miss updates, and HR staff struggle to manage everything manually.

Smart Care System is developed to solve these problems. It is a smart digital platform that helps caretaker companies manage their services easily. The system supports everything from assigning caretakers, tracking their work, approving leave requests, handling complaints, to collecting feedback from clients.

The system connects Admins, HR Managers, Caretakers, and Clients in one place. It saves time, reduces errors, improves trust, and makes home care services more organized and reliable for everyone involved.

1.2. Proposed Solution

To solve the problems in the current way caretaker services are managed, we suggest creating a Smart Care System (SCS). This system will make everything easier by using technology. The SCS will:

1. One Place for Everyone to Work Together:

Bring Admins, HR staff, Caretakers, and Clients into one system so they can easily share information and work together.

2. Make Tasks Automatic:

Automatically handle things like assigning caretakers, tracking their work hours, approving leave, and managing their schedules. This will save time and avoid mistakes.

3. Help Everyone Communicate Easily:

Make it simple for caretakers and clients to talk to each other, share updates, send

complaints, and give feedback.

4. Create Useful Reports:

The system will show reports about caretaker work, client feedback, and attendance. This helps improve the quality of service over time.

1.3. How is it different in the Modern Era?

In the modern era, digital transformation has become essential for managing day-to-day operations across all industries, including the home care sector. However, many caretaker companies still rely on manual processes, phone calls, and paper-based records, which lead to errors, delays, and miscommunication.

The Smart Care System (SCS) introduces a modern, technology-driven approach to managing caretaker services. Unlike traditional methods, SCS offers:

1. Centralized Digital Platform:

All users — Admins, HR Managers, Caretakers, and Clients — interact through a single, unified system. This eliminates scattered information and enables real-time collaboration and transparency.

2. Automation of Core Tasks:

Routine operations like caretaker assignment, schedule creation, leave approvals, and complaint tracking are automated. This reduces the dependency on human coordination and ensures faster, more accurate task execution.

3. Smart Communication Features:

Clients and caretakers can communicate through built-in messaging, complaint submission, and feedback options. This ensures that no important updates are missed and builds trust and accountability.

4. Data-Driven Insights:

The system generates real-time reports on attendance, performance, and client satisfaction, enabling companies to make informed decisions and continuously improve their services.

5.Remote Access & Cloud-Based Operation:

Being a web-based system, SmartCare can be accessed from anywhere, which suits the remote and mobile nature of caretaker services in today's connected world.

02. Project Goal

The main goal of the Smart Care System is to make caring for elderly people, children, or patients easier and more reliable. It aims to connect families, caretakers, and administrators through one simple platform that helps with scheduling visits, sharing updates, and tracking care activities in real time. This system will reduce stress, improve communication, and ensure better care for those who need it.

03. Project Scope

3.1. In-Scope (What the System Will Cover)

The SmartCare system will include the following core functionalities:

- **User Management:**
 - Admin can register HR Managers and Caretakers.
 - Clients can self-register via the system.
- **Caretaker Scheduling:**
 - HR can assign caretakers to clients based on service needs and availability.
 - System generates customized caretaker schedules.
- **Leave Management:**
 - Caretakers can apply for leave.
 - HR/Admin can approve or reject leave requests.
- **Complaint Management:**
 - Clients can submit complaints.
 - HR can view, respond to, or escalate complaints.
- **Feedback System:**
 - Clients can provide service feedback for caretakers.
- **Priority Tagging:**
 - HR can suggest and Admin can assign high-priority status to caretakers.
- **Service Management:**
 - Admin can define and manage different care services (e.g., elder care, cleaning, babysitting).

- **Reports & Dashboards:**
 - Admin and HR can view system-generated reports (attendance, feedback, salary logs, etc.).
- **Login and Authentication:**
 - Role-based secure login for Admin, HR, Caretakers, and Clients.
 - Forgot/Reset password feature.
- **Basic UI/UX:**
 - Simple and user-friendly interface accessible by all types of users.
- **Payment Mechanism:**
 - Clients can make online payments for services through secure payment gateways.
 - The system records payments and tracks their status.

3.2. Out-of-Scope (What the System Will Not Cover)

The following functionalities are intentionally excluded from this version of the system:

- **Live chat or video calling** features.
- **Mobile app** version (web only for now).
- Integration with third-party tools like payroll or HRMS systems.
- **Multi-company support** — the system is built for one caretaker company only.
- Managing **school or property caretakers** (only elder care, babysitting, and maid services are supported).

3.3. Project Boundaries

- Designed for a **single caretaker company** with a fixed set of services.
- Web-based system accessible through standard browsers.
- The system will support **four user roles**: Admin, HR Manager, Caretaker, Client.
- Intended for **internal and client-side use**; not designed for public marketplace listing.

3.4. Constraints

- Limited to a **9-month development period** by a team of 4 members.
- The development environment is constrained to the following technologies:
 - **Frontend:** HTML, CSS, JavaScript
 - **Backend:** PHP
 - **Database:** MySQL
 - **Web Server:** Apache
 - **Version Control:** Git/GitHub
- The system is expected to run on a **moderate-performance hosting server**.
- UI will be designed to be **functional but simple**, with minimal animations or advanced visuals.

3.5. Assumptions

- The caretaker company has internet access and basic computer infrastructure.
- End users (clients and caretakers) have **basic digital literacy**.
- The company has a **fixed set of service types** that do not frequently change.
- Feedback and complaints will be provided **in good faith and without misuse**.
- HR and Admin will be trained to use the system effectively.
- No legal restrictions prevent storing basic personal data like name, contact, and service logs.

04. Objectives of the project

The primary objective of the SmartCare – Caretaker Management System is to design and implement a comprehensive, efficient, and user-friendly web-based platform that connects clients (such as elders or families in need of domestic care) with professional caretakers, maids, and babysitters, while streamlining administrative and operational processes for the caretaker company.

4.1. General Objective

To develop a centralized caretaker management system that simplifies service assignment, user role management, schedule tracking, leave handling, and profile maintenance, enhancing the overall caregiving experience for both clients and the caretaker company.

4.2. Specific Objectives

1. Implement Role-Based Login System

Develop a secure login and access control system for multiple user roles: Admin, HR Manager, Caretaker, Maid, Babysitter, and Client. Each role will access a personalized dashboard with role-specific features.

2. Caretaker Profile Management

Enable caretakers to view and update their personal and professional profiles, including contact details, specialization, experience, and availability.

3. View Assigned Clients and Schedule

Provide caretakers with an organized view of their upcoming tasks, assigned clients, service type, and time slots for better planning and efficiency.

4. Leave Request and Approval System

Allow caretakers to request leave by submitting dates and reasons. HR managers can approve or reject leave requests based on caretaker availability and schedule conflicts.

5. Client Registration and Service Booking

Allow clients to register, log in, and request services like elder care, babysitting, or housekeeping based on preferred date, time, and caretaker type.

6. Caretaker Assignment

Enable the HR/Admin to assign caretakers either manually or automatically based on availability, client preferences, and geographic proximity.

7. Customized Caretaker Schedule Creation

Allow HR managers to manually create or adjust individual caretaker schedules (e.g., fixed clients on specific days or shifts), overriding automatic allocations if needed.

8. Location-Based Filtering

Match clients with nearby caretakers to optimize travel time and ensure timely service. This helps in balancing assignments across caretakers.

9. Service History Tracking

Maintain detailed logs of each caretaker's service history, including date, service type, client, and feedback received.

10. High-Priority Caretaker Tagging

Allow admins or HR managers to tag reliable or top-performing caretakers as high-priority for handling urgent or clients who have been using the service for over a specific period (VIP clients).

11. Complaint Management System

Allow clients to file complaints against services or caretakers. HR/Admin can view, track, and resolve these complaints, maintaining transparent logs for quality assurance.

12. Reporting and Analytics for Admin

Generate monthly or custom reports including service volume, caretaker utilization, leave reports, complaint statistics, and client feedback summaries.

13. Client Feedback System

Collect structured feedback from clients after each service, which helps improve caretaker performance and service quality monitoring.

14. Notifications and Reminders

Send automatic alerts to caretakers and clients regarding upcoming service appointments, leave decisions, or schedule changes.

15. Service Status Updates

Allow caretakers to mark services as completed, and allow clients to confirm or rate the completed service for accountability and transparency.

16. Security and Access Control

Protect all user data using encrypted passwords, session-based login handling, and strict access based on user roles.

05. Project Feasibility

5.1. Technical Feasibility

Category	Technology/ Tool	Short Description
Frontend	HTML	Structures the web pages and content.
	CSS	Styles the layout and ensures responsive design.
	JavaScript	Adds interactivity and client-side logic.
Backend	PHP	Handles server-side logic like form processing, sessions, and validations.
	MySQL	Stores user data, schedules, payments, and history.
Web Server	Apache HTTP Server	Hosts the web application and supports PHP and MySQL integration.
Local Testing	XAMPP / WAMP	Bundled tools for local development with Apache, PHP, and MySQL.
Version Control	Git	Tracks code changes, enables collaboration, and version management.
Code Repository	GitHub	Stores project code, supports collaboration and issue tracking.
Documentation	Google Docs	Used for writing reports, proposals, and shared team documents.
Project Management	Jira	Tracks tasks, sprints, and project progress.
Prototyping	Figma	Designs UI mockups and wireframes with real-time team collaboration.

5.2. Economic Feasibility

Economic feasibility examines whether the system's benefits justify its costs. SmartCare is expected to improve efficiency, reduce manual tasks, and offer long-term savings for the caretaker company.

5.2.1. Cost Analysis

- **Development Costs**
 - **Labor:** No paid developers; student hours simulate real-world effort.
 - **Software Tools:** Free/open-source (VS Code, Git, Figma, MySQL, XAMPP).
 - **Hardware:** Personal laptops and internet—no additional hardware needed.
- **Hosting & Deployment Costs**
 - **Web Hosting:** Optional; LKR 5,000–10,000/year on platforms like Hostinger.
 - **Domain Name:** Approx. LKR 2,000–3,000/year.
 - **Local Deployment:** Free via XAMPP for academic/demo purposes.
- **Maintenance & Upgrades**
 - Minimal costs; occasional bug fixes or feature updates may require developer effort.
- **Training & Support**
 - No formal training cost; time allocated for user manuals and demos for Admin, HR, and Caretakers.

5.2.2. Benefit Analysis

- **Operational Efficiency:**

- Automates bookings, assignments, leave tracking—saving HR time and reducing paperwork.
- **Improved Accuracy:**
 - Reduces scheduling and payroll errors through automation.
- **Better Service Quality:**
 - Faster client responses and balanced workloads enhance client satisfaction.
- **Scalability:**
 - System supports future growth (more users/services) at low additional cost.
- **Long-Term Savings:**
 - Reduces reliance on physical records and manual coordination, cutting recurring costs.

5.3. Social Feasibility

Social feasibility evaluates whether the proposed SmartCare – Caretaker Management System will be accepted by its intended users, including caretakers, HR managers, administrators, and clients. It also considers the system’s potential to positively impact users’ daily activities and improve coordination in the caregiving process.

5.3.1. Assumptions and User Expectations

Although no formal surveys or interviews were conducted due to time and resource constraints, the system's requirements were identified through:

- Informal discussions with peers and individuals familiar with caretaker and service-based work environments.
- Observations of general industry challenges such as manual scheduling, inefficient leave handling, and lack of centralized tracking.
- Secondary research through online platforms and comparisons with existing service management tools.

From these sources, several user needs were identified:

- Caretakers require a system that shows their assignments and allows them to manage their leave and profile.
- Clients need transparency in who provides services and a way to view service history or give feedback.
- HR/Admin staff benefit from automation in assigning staff, managing leaves, and generating reports.

5.3.2. Anticipated Social Benefits

The SmartCare system is designed to address these concerns and offer social value across all stakeholder roles:

1. For Caretakers

- a. Ability to track schedules, apply for leave, and update their profiles with ease.
- b. Increased clarity around assignments and salary tracking.

2. For Clients

- a. Clear information about assigned caretakers and ability to review service history.
- b. Higher confidence in service reliability and quality.

3. For Admin/HR

- a. Streamlined workflow for managing staff schedules and leaves.
- b. Fewer manual errors and reduced paperwork.

5.3.3. User Interface Accessibility

One of the core goals of the system is inclusivity and ease of use. The SmartCare platform features a **simple and intuitive user interface** designed for users with **varying levels of digital literacy**. Icons, clear navigation menus, and clean page layouts ensure that caretakers, even with limited technical knowledge, can easily learn and use the system. This greatly increases the likelihood of system acceptance and long-term usage across all user groups.

5.4. Operational Feasibility

Operational feasibility checks if the SmartCare system can be effectively used and sustained in real caretaker company environments.

5.4.1. Integration with Existing Operations

- Designed to replace manual methods with digital workflows.
- Automates caretaker assignments, leave tracking, and schedules.
- Eases HR workload and improves coordination and record-keeping.

5.4.2. User Adaptability and Training

- Simple interface with clear navigation and role-specific dashboards.
- Minimal training needed; user guides or demo videos can assist.
- **Caretakers:** View schedules, apply for leave, track work.
- **HR/Admins:** Assign staff, manage users, generate reports.
- **Clients:** Request services, view history, give feedback.

5.4.3. Maintenance and Support

- Built using common tools (PHP, MySQL) and easy to maintain.
- Companies can use internal IT or third-party help for:
 - Backups, security, feature updates, scaling.
- Future support may include bug fixes, UI upgrades, and new features.

5.4.4. Scalability and Long-Term Use

- Easily adaptable to new caretaker roles or services.
- Can scale to more users, locations, and features as the business grows.
- Enables advanced reporting as data increases over time.

5.5. Legal and Ethical Feasibility

5.5.1. Legal Feasibility

Legal feasibility assesses whether there are any **laws, regulations, or policies** that could affect the development, deployment, or use of the SmartCare system.

1. Data Protection and Privacy

SmartCare will store sensitive data, including:

- Caretaker details (name, contact, availability, etc.)
- Client information (addresses, service needs)
- Salary and service history records

Although this project is for academic purposes, in a real-world deployment, it must comply with:

- **Personal Data Protection Act (Sri Lanka)** – if applicable
- **General principles of privacy laws** such as:
 - Not collecting more data than necessary
 - Storing data securely
 - Limiting data access based on user roles
 - Not sharing personal data without consent

2. Compliance with IT Regulations

- The system will not involve illegal content, third-party copyright violations, or harmful features.
- The technologies used (PHP, MySQL, GitHub, etc.) are all open-source or licensed for educational and commercial use.

5.5.2. Ethical Feasibility

Ethical feasibility focuses on ensuring that the system is designed and used in a **morally responsible** way and promotes **fair treatment, transparency, and trust** among its users.

1. User Consent and Transparency Our team consists of **four members**, each with prior experience in web development, UI/UX design, and system architecture. This collective skill set enables the team to manage the project's complexity efficiently while maintaining development quality.

- Only authorized users (based on roles) can access specific parts of the system.
- Users are aware of what information is stored and how it will be used.
- Clients and caretakers will have access to their own data.

2. Fairness and Equal Access

- No user is given unfair advantage; all roles have clearly defined permissions.
- The system does not discriminate based on gender, religion, or age.

3.Data Accuracy and Responsibility

- Admins and HR managers are expected to update and verify information correctly.
- The system promotes accountability (e.g., through service history tracking and feedback).

4.Digital Literacy Consideration

- The system is designed with a simple and accessible interface to ensure that even users with basic digital skills can use it effectively.

5.6 Schedule Feasibility

The SmartCare – Caretaker Management System is planned with a strong emphasis on schedule feasibility. Our team is utilizing the **iterative development model**, as the core requirements are well-defined but the model allows flexibility to revise and enhance features as feedback is received during development. This approach ensures continuous improvement without derailing the project timeline.

The total duration allocated for the project is **nine months**, with **eight months dedicated to core development activities** and one month allocated for testing, documentation, and final adjustments. This timeline is realistic and allows adequate time for planning, design, implementation, testing, and refinements.

Considering these factors, we can confidently conclude that the schedule feasibility of the SmartCare project is **high**. With a structured approach, dedicated team members, and an iterative model, the project is on track to meet its completion deadline without compromising quality.

5.6.1. Man-Hour Calculation

- **Estimated working time per member per week** = 10 hours
- **Estimated duration** = 40 weeks
- **Total hours per member** = 10 hours $\times$ 40 weeks = **400 hours**
- **Team size** = 4 members
- **Estimated total man-hours for the project** = 400 hours $\times$ 4 members = **1600 hours**

This allocation ensures enough time is available for requirements gathering, UI/UX design, frontend and backend development, integration, testing, and documentation. It also accommodates minor delays or scope adjustments within the project timeline.

06. Deliverables of the project

- **Fully Functional Web – Based Application**

A responsive and user friendly web application that includes all core features such as user registration, caretaker listings, service booking, feedback collection, and administrative controls.

- **Role-Based Dashboards**

To provide each user type (Admin, HR Manager, Client, and Caretaker) with a personalized and secure interface that displays only the features, information and tools relevant to their specific role.

- **User Documentation**

A comprehensive user guide for clients, caretakers, HR managers and administrators explaining how to use the system effectively.

- **Source Code Repository**

The complete source code of the system, properly structured and commented, delivered via a version-controlled platform.

- **Complete Documentation of the entire project**

Includes all essential documents such as the System Requirements Specification (SRS) and User Manual that describe the design, functionality, usage and maintenance of the system.

07. Project Constraints and Assumptions

7.1. Project Constraints

- Each user(Admin, HR Manager, Client, Caretaker) can only access features relevant to their role. Unauthorized access to other modules is strictly prohibited.
- Caretakers can only be booked if they are marked as available in the system. Overlapping or double-booking of services is not permitted.
- Service bookings must be made at least 24 hours in advance to allow for caretaker confirmation and scheduling.
- Sensitive user data is only visible to authorized parties and must be securely stored and transmitted.
- Clients can only view caretakers based on their selected city, district or postal code. Cross-location bookings are not allowed.

7.2. Project Assumptions

- The system will be developed using commonly available web technologies (HTML, CSS, JS, PHP, MySQL) and deployed on a standard hosting environment.
- All users will have stable internet access to interact with the system, as it is a web-based platform.
- All users (Admin, HR Manager, Clients, and Caretakers) must be registered in the system with valid credentials before accessing features.
- All Caretaker profiles, availability status, and client information are correctly entered and maintained in the database.
- Users will access the system using one device/session at a time to avoid session conflicts.

- Sufficient development resources (time, technical expertise, and hardware/software tools) are available to complete the project within the proposed timeline.

08. Requirements

8.1. Functional Requirements

8.1.1. Functionalities

8.1.1.1. Admin(Caretaker Company)

- Login
- View dashboard with stats on clients, caretakers, bookings, and service usage
- Manage Users (HR Manager, Caretakers and Clients)
- View all users with filters by role or service type
- Manage client feedback and flag inappropriate reviews
- View and export reports (usage, payments, performance)
- View caretaker leave records
- Monitor full history logs and user activity
- Send system-wide announcements or notifications
- View payment reports and financial summaries

8.1.1.2. HR Manager

- Login
- Manage caretaker work schedules
- Approve or decline caretaker leave requests
- Assign caretakers to client requests/services
- Track caretaker attendance and availability
- View caretaker performance based on client feedback
- View service history for each caretaker
- Generate reports on caretaker activities and services
- Assign or remove high-priority status to caretakers
- Adds or update caretaker working locations (city/district/postal code)

- Approve or reject client service requests
- Handle complaints related to caretakers or services

8.1.1.3. Caretaker (Elder caretakers / Maids / Babysitters)

- SignUp
- Login
- View and update personal profile details
- View assigned client list and daily/weekly work schedule
- Apply for leave with reason and preferred dates
- View leave request status and complete leave history
- View feedback and ratings received from clients
- Access service history (past appointments, hours worked, service types)
- Mark attendance/check-in and check-out for client visits
- View personal priority status and badge
- Receive priority or special bookings based on status/performance
- Get notifications for new request, updates or changes in schedule
- Accept or reject assigned service requests within allowed time frame
- Report issues or complaints related to clients or service tasks

8.1.1.4. Client (Elder or Family)

- SignUp
- Login
- View available caretakers with profile information, ratings & comments, real-time service availability
- Book services by selecting caretaker, preferred date and time
- Make an advance payment when requesting a caretaker service
- Remaining balance is paid after the service is completed
- Give feedback & rating after service
- Complaining about caretakers
- View booking history and service details
- View payment history
- Cancel or reschedule contract
- Contact the assigned caretaker

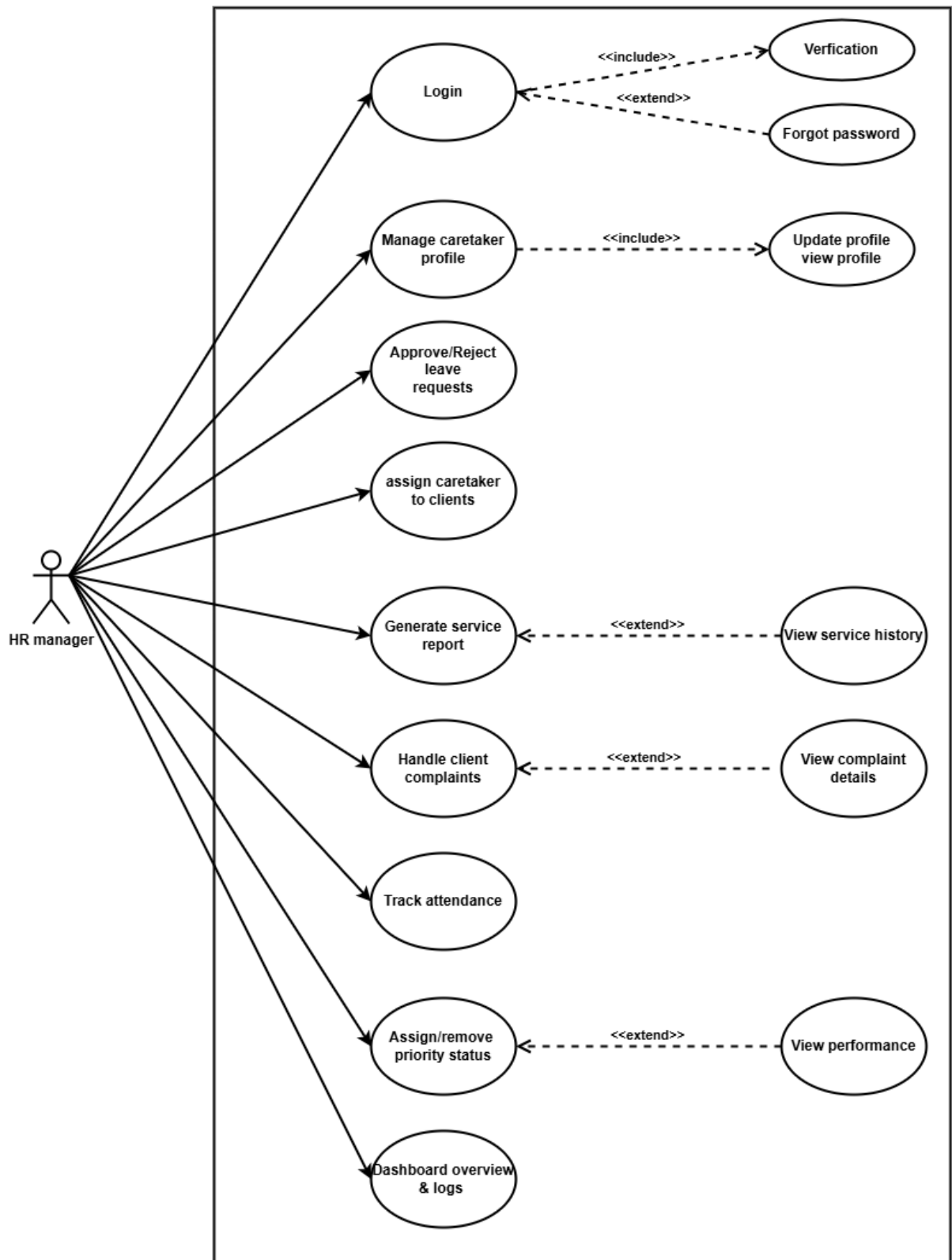
- Change assigned caretaker if not satisfied
- Create customized service plans for caretakers
- Filter caretakers by location

8.1.2. Use Case Diagrams

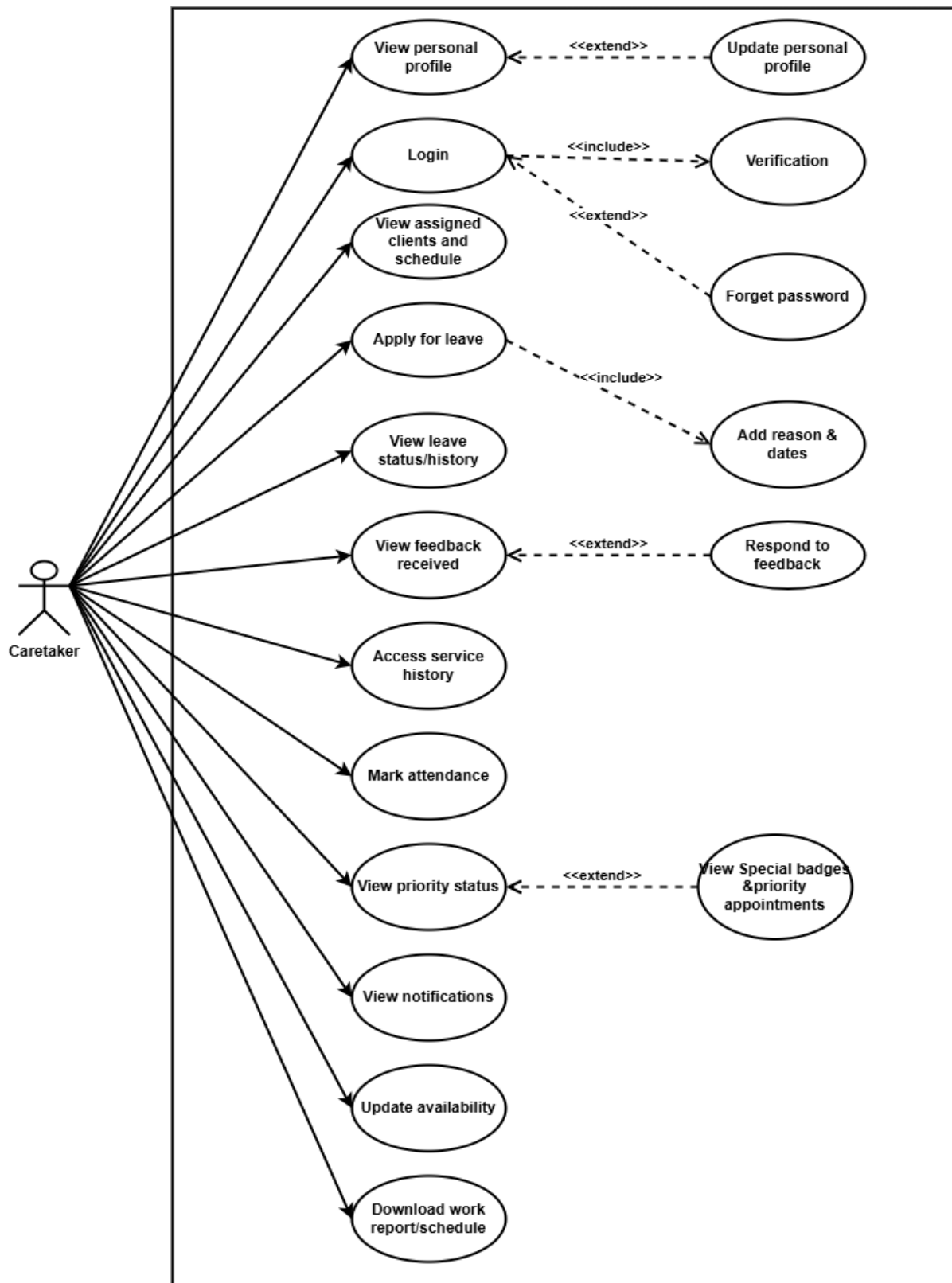
8.1.2.1. Admin



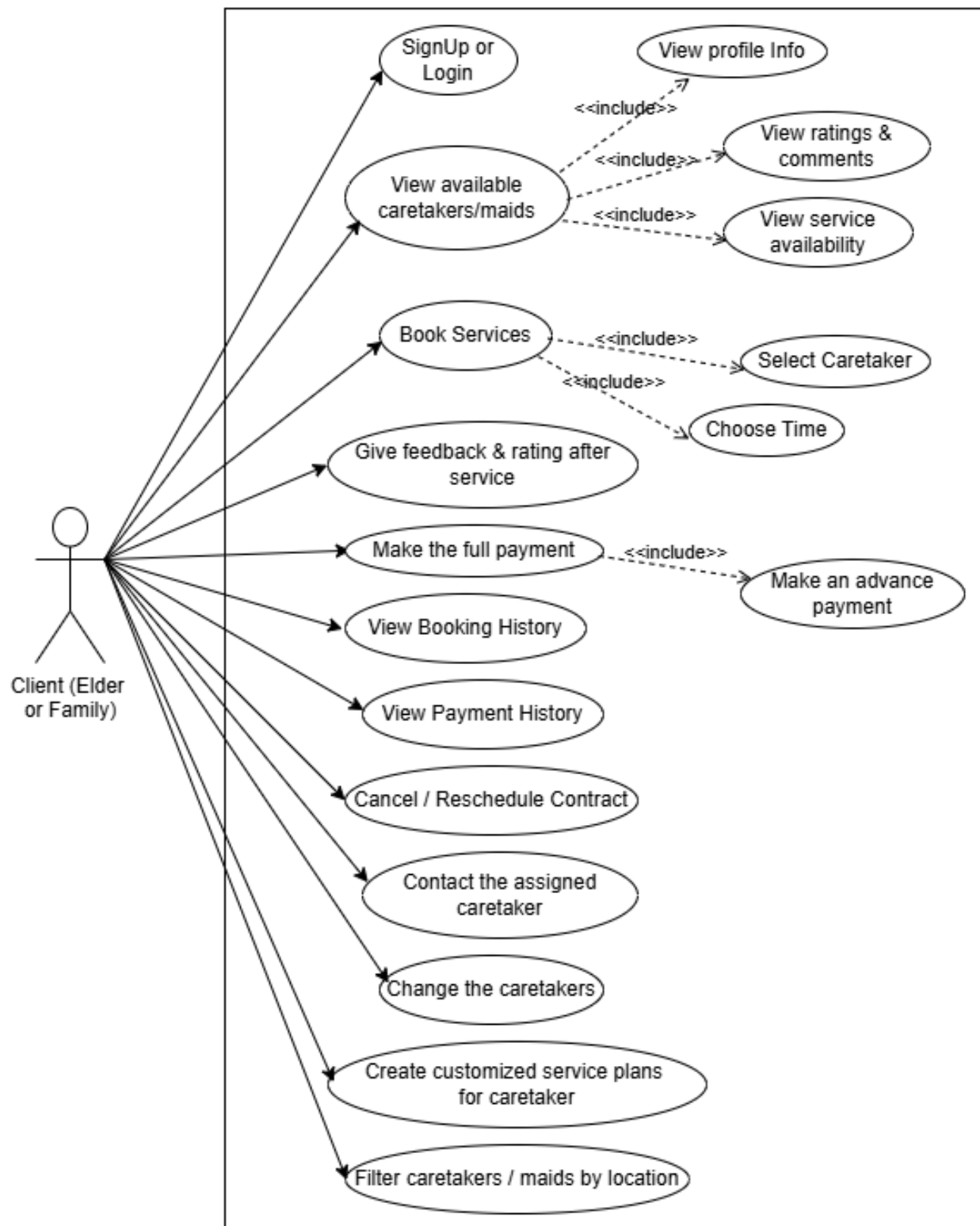
8.1.2.2. HR Manager



8.1.2.3. Caretaker



8.1.2.4. Client



8.1.3. Use case Narratives

8.1.3.1. Common use cases

Use Case ID	CUC-01
Use Case Name	Login
Primary Actor/s	Admin, HR Manager, Caretaker, Client
Description	Allows a registered user to log in using their credentials.
Pre-conditions	User has a valid, verified account.
Main Scenario	• User navigates to login page • Enter email and password. • Click “Login”. • System validates credentials. • User is redirected to the respective dashboard.
Includes	<<include>> Verification
Extensions	<<extend>> Forgot Password
Exceptions	• Invalid email or password. • Account not verified.
Post-conditions	Users are successfully logged in and can access their dashboard.

Use Case ID	CUC-02
Use Case Name	Verification
Primary Actor/s	Any Registered User
Description	Confirms the user's identity using email verification during registration or login.
Pre-conditions	A valid registration attempt.
Main Scenario	• User receives a verification email. • Click on the link. • System marks account as verified.
Exceptions	• Link expired. • Already verified.
Post-conditions	Users can now log in.

Use Case ID	CUC-03
Use Case Name	Forgot Password
Primary Actor/s	Admin, HR Manager, Caretaker, Client
Description	Allows users to reset their password if forgotten.
Pre-conditions	Users must have a registered email.
Main Scenario	• Click "Forgot Password". • Enter email address. • System sends a reset link. • User clicks a link and enters a new password. • Password is updated.
Exceptions	• Email doesn't exist → show error. • Link expired.
Post-conditions	Password updated, user can log in.

Use Case ID	CUC-04
Use Case Name	Logout
Primary Actor/s	Any Logged-in User
Description	Terminates the current user session.
Pre-conditions	Users must be logged in.
Main Scenario	• User clicks “Logout” button • Session is terminated. • User is redirected to the login page.
Post-conditions	Session is cleared, and the user is logged out.

Use Case ID	CUC-05
Use Case Name	Update Personal Profile
Primary Actor/s	All Users
Description	Allows a user to update their profile information such as name, phone, or photo.
Pre-conditions	Users must be logged in.
Main Scenario	• Navigates to profile. • Click “Edit”. • Updates fields. • Submits. • System saves changes.
Exceptions	• Invalid phone/email format.
Post-conditions	Profile updated.

8.1.3.2. Admin Use Case

Use Case ID	A-01
Use Case Name	Login
Primary Actor/s	Admin
Description	Allows Admin to log in to the system.
Pre-conditions	Admin has a verified account.
Main Scenario	• Admin enters credentials • System validates and logs in users.
Post-conditions	Admin is logged in and redirected to the dashboard.
Exceptions	Invalid credentials or unverified account.

Use Case ID	A-02
Use Case Name	Logout
Primary Actor/s	Admin
Description	Logs the Admin out of the system securely.
Pre-conditions	Admin is logged in.
Main Scenario	• Admin clicks logout. • Session ends.
Post-conditions	Admin is logged out.
Exceptions	Session timeout or server error.

Use Case ID	A-03
Use Case Name	Manage Users
Primary Actor/s	Admin
Description	Allows Admin to manage (add/edit/delete) user accounts.
Pre-conditions	Admin is logged in.
Main Scenario	• Admin views user list. • Registers/edits/deletes users.
Includes	Register Caretakers Register HR Manager
Post-conditions	User records updated successfully.
Exceptions	User not found, database error.

Use Case ID	A-04
Use Case Name	Generate Reports
Primary Actor/s	Admin
Description	Generates various operational and service reports.
Pre-conditions	Data exists in the system.
Main Scenario	• Admin selects report type/date. • System generates reports.
Post-conditions	Report is available for viewing or download.
Exceptions	No data available.

Use Case ID	A-05
Use Case Name	Manage Services
Primary Actor/s	Admin
Description	Allows Admin to add/update/delete service records.
Pre-conditions	Admin is logged in.
Main Scenario	Admin edits or adds new service categories and details.
Post-conditions	Services list is updated.
Exceptions	Invalid input or service ID not found.

Use Case ID	A-06
Use Case Name	Manage Payment Handling
Primary Actor/s	Admin
Description	Manages client and caretaker payment records.
Pre-conditions	Payment data exists.
Main Scenario	Admin views/updates payment entries.
Post-conditions	Payments are processed or marked.
Exceptions	Payment gateway failure or record mismatch.

Use Case ID	A-07
Use Case Name	View Leave Requests
Primary Actor/s	Admin
Description	Allows Admin to review leave requests submitted by staff.
Pre-conditions	Leave request data is present.
Main Scenario	• Admin views requests • Take action if needed.
Post-conditions	Request is marked as viewed.
Exceptions	

	No leave requests found.
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Use Case ID	A-08
Use Case Name	View Dashboard
Primary Actor/s	Admin
Description	Displays system-wide statistics and summaries to the Admin.
Pre-conditions	Admin is logged in.
Main Scenario	Admin sees system overview on dashboard load.
Post-conditions	Summary is visible.
Exceptions	Dashboard fails to load or shows incomplete data.

Use Case ID	A-09
Use Case Name	View Feedback/Rating
Primary Actor/s	Admin
Description	Admin can view feedback and ratings submitted by clients.
Pre-conditions	Feedback has been submitted.
Main Scenario	Admin views rating comments and scores.
Post-conditions	Feedback is read or exported.
Exceptions	

	No feedback available.
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8.1.3.3. HR Manager Use cases

Use Case ID	HR-01
Use Case Name	Manager/Caretaker Profile
Primary Actor	HR Manager
Description	Allows HR Manager to view and manage manager and caretaker profiles.
Pre-conditions	HR Manager is logged in.
Main Scenario	HR Manager navigates to profile management. Views, updates, or removes profile details.
Post-conditions	Profiles are successfully viewed or updated.
Exceptions	Profile not found.

Use Case ID	HR-02
Use Case Name	Approve/Reject Leave Request
Primary Actor/s	HR Manager
Description	Allows the HR Manager to approve or reject leave requests submitted by staff.
Pre-conditions	Leave request has been submitted.
Main Scenario	HR Manager views pending leave requests. Approves or rejects with comments.
Post-conditions	Leave request status is updated.
Exceptions	Invalid request ID or system error.

Use Case ID	HR-03
Use Case Name	Assign Caretaker to Client
Primary Actor/s	HR Manager
Description	Assigns a caretaker to a specific client.
Pre-conditions	Client and caretaker profiles are active.
Main Scenario	HR Manager selects clients. Chooses available caretaker. Assigns caretaker.
Post-conditions	Assignment is successful.
Exceptions	No available caretaker or system error.

Use Case ID	HR-04
Use Case Name	Generate Service Report
Primary Actor/s	HR Manager
Description	Generates detailed reports of caretaker services.
Pre-conditions	Caretaker has service data.
Main Scenario	HR Manager selects date range and caretaker. System generates reports.
Includes	◊ View Service History
Post-conditions	Report is generated and viewable.
Exceptions	No data or access denied.

Use Case ID	HR-05
Use Case Name	Handle Client Complaints
Primary Actor/s	HR Manager
Description	Handles complaints submitted by clients.
Pre-conditions	Complaint has been submitted.
Main Scenario	HR Manager views complaints. Take action.
Extensions	View Complaint Details
Post-conditions	Complaints are resolved or escalated.
Exceptions	Complaint not found.

Use Case ID	HR-06
Use Case Name	Track Attendance
Primary Actor/s	HR Manager
Description	Tracks attendance of caretakers and staff.
Pre-conditions	Attendance is being logged.
Main Scenario	HR Manager views the attendance dashboard. Filters by employee or date.
Post-conditions	Attendance details are visible.
Exceptions	Data not available.

Use Case ID	HR-07
Use Case Name	Assign/Remove Priority Status
Primary Actor/s	HR Manager
Description	Assigns/removes priority status to caretakers based on performance.
Pre-conditions	Caretaker profile exists.
Main Scenario	HR Manager views the caretaker profile. Assigns/removes priority.
Extensions	View Performance
Post-conditions	Priority status updated.
Exceptions	System error or caretaker not found.

Use Case ID	HR-09
Use Case Name	View Logs
Primary Actor/s	HR Manager
Description	Views activity logs for auditing.
Pre-conditions	Log data is stored.
Main Scenario	The HR Manager opens the log section. Filters and views entries.
Post-conditions	Logs are displayed for review.
Exceptions	No logs or access denied.

8.1.3.4. Caretaker Use cases

Use Case ID	CT-01
Use Case Name	View Personal Profile
Primary Actor	Caretaker
Description	Allows caretakers to view their profile details including name, email, contact, and job role.
Pre-conditions	Caretaker must be logged in.
Main Scenario	• Caretaker logs in. • Navigates to "My Profile". • System displays personal profile details.
Exceptions	• System fails to load data → error shown.
Post-conditions	Profile information is displayed.

Use Case ID	CT-02
Use Case Name	View Assigned Clients & Schedule
Primary Actor	Caretaker
Description	Allows caretakers to view their assigned clients along with the dates and time slots for service.
Pre-conditions	Caretaker must be logged in.
Main Scenario	• Logs in. • Open the schedule tab. • System fetches and displays client details and timing.
Exceptions	• No clients assigned → show notification.
Post-conditions	Daily/weekly schedule is shown to the caretaker.

Use Case ID	CT-03
Use Case Name	Apply for Leave
Primary Actor	Caretaker
Description	Enables caretakers to request leave by submitting a reason and duration.
Pre-conditions	Caretaker must be logged in.
Main Scenario	• Open leave form. • Fills in reason and date range. • Submit the request.
Exceptions	• Invalid date range → error shown.
Post-conditions	Leave request is submitted for HR review.

Use Case ID	CT-04
Use Case Name	View Leave Status / History
Primary Actor	Caretaker
Description	Allows caretakers to view the approval status and history of all their leave requests.
Pre-conditions	Caretaker must have submitted leave requests.
Main Scenario	• Logs in. • Opens leave the history section. • System shows leave records and their status.
Exceptions	• No records → "No leave applied yet" shown.
Post-conditions	Leave history and current status displayed.

Use Case ID	CT-05
Use Case Name	View Feedback Received
Primary Actor	Caretaker
Description	Displays feedback provided by clients after completed service sessions.
Pre-conditions	Feedback must exist.
Main Scenario	• Open feedback module. • System displays client ratings and comments.
Exceptions	• No feedback yet → "No feedback available" message.
Post-conditions	Feedback is shown.

Use Case ID	CT-06
Use Case Name	Access Service History
Primary Actor	Caretaker
Description	Lets caretakers view details of previously completed services.
Pre-conditions	Service sessions must exist.
Main Scenario	• Logs in. • Navigates to "Service History". • Views list of completed tasks with dates.
Post-conditions	Service records are visible.

Use Case ID	CT-07
Use Case Name	Mark Attendance
Primary Actor	Caretaker
Description	Records attendance by allowing caretakers to check in/out for assigned shifts.
Pre-conditions	Caretaker must be logged in and scheduled for a shift.
Main Scenario	• Click “Mark Present” or “Check In”. • System stores timestamps and shift data.
Post-conditions	Attendance is saved.

Use Case ID	CT-08
Use Case Name	View Priority Status
Primary Actor	Caretaker
Description	Shows whether a caretaker holds priority assignments or badges.
Pre-conditions	The caretaker must be logged in and assigned a priority status by the HR/Admin.
Main Scenario	• Open priority status page. • System displays labels like "High Priority", "Experienced", etc.
Post-conditions	Status details are shown.

Use Case ID	CT-09
Use Case Name	View Notifications
Primary Actor	Caretaker
Description	Displays real-time messages, alerts, and updates related to shifts, approvals, and client feedback.
Pre-conditions	The caretaker must be logged into the system and have active notifications available in their account.
Main Scenario	• Open notification panel. • Read messages or updates.
Post-conditions	Notifications are marked as read.

Use Case ID	CT-10
Use Case Name	Update Availability
Primary Actor	Caretaker
Description	Let caretakers inform HR of their available dates/times for service.
Pre-conditions	The caretaker must be logged into the system and not currently on approved leave.
Main Scenario	• Open availability form. • Choose dates and time slots. • Submit form.
Post-conditions	System updates the caretaker's availability calendar.

Use Case ID	CT-11
Use Case Name	Download Work Report/Schedule
Primary Actor	Caretaker
Description	Lets caretakers export their schedule or past service data as PDF/Excel.
Pre-conditions	The caretaker must be logged into the system and have completed service records or scheduled assignments available.
Main Scenario	• Open report section. • Select date range. • Click “Download”.
Post-conditions	Report is downloaded successfully.

8.1.3.5. Client Use cases

Use Case ID	C-01
Use Case Name	Sign Up
Primary Actor/s	Unregistered client
Description	The client registers a new account using their credentials
Pre-conditions	The client is not logged in
Main Scenario	- Client navigates to the registration page. - Enter required credentials (email, password). - Fills in personal details and submits the form. - System verifies credentials or sends a verification link. - Upon success, the client is redirected to the dashboard.
Exceptions	- If login fails, the system displays an error. - If an email is already registered during signup, the system prompts for a new email. - If email is not verified, access is restricted.
Post-conditions	The client is successfully registered and redirected to the platform dashboard.

Use Case ID	C-02
Use Case Name	Login
Primary Actor/s	Registered client
Description	The client logs into the system using their credentials
Pre-conditions	The client is not logged in
Main Scenario	- Client navigates to the login page - Enter required credentials (email, password) - Submits the login form - System verifies credentials or sends a verification link - Upon success, the client is redirected to the dashboard
Exceptions	-If login fails, the system displays an error
Post-conditions	The client is successfully logged in and redirected to the platform dashboard.

Use Case ID	C-03
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Use Case Name	View Available Caretakers
Primary Actor/s	Client (Elder or Family)
Description	The client views a list of available elder caretakers, maids, or babysitters.
Pre-conditions	The client is logged into the system.
Main Scenario	- The client navigates to the "Available Caretakers" section. - The system displays a list of caretakers based on availability. - Clients can click on a profile to view detailed information. - Clients can apply filters by city, district, or postal code.
Exceptions	- If no caretakers are available in the selected area, the system displays a relevant message.
Post-conditions	The client views a filtered list of available caretakers and can proceed to book a service.

Use Case ID	C-04
Use Case Name	View Profile Informations
Primary Actor/s	Client
Description	The client views detailed profile information of a selected caretaker
Pre-conditions	The client is logged in and has accessed the list of available caretakers.
Main Scenario	- The client selects a caretaker from the list. - The system displays the caretaker's profile, including personal details, experience, and service types.
Exceptions	- If the profile data fails to load, an error message is shown
Post-conditions	The client successfully views the caretaker's profile.

Use Case ID	C-05
Use Case Name	View Ratings & comments
Primary Actor/s	Client
Description	The client views ratings and feedback left by other clients for a caretaker.
Pre-conditions	The client is viewing a caretaker's profile.
Main Scenario	- Client scrolls to the feedback section. - System displays previous ratings and comments from other clients.
Exceptions	- If no feedback is available, the system shows a "No feedback yet" message.
Post-conditions	Client gains insights about the caretaker's service quality.

Use Case ID	C-06
Use Case Name	View Service Availability
Primary Actor/s	Client
Description	The client checks when a selected caretaker is available for booking.
Pre-conditions	The client is viewing a caretaker's profile.
Main Scenario	- System displays the caretaker's weekly/monthly availability schedule. - Client reviews free time slots.
Exceptions	- If availability cannot be retrieved, an error message is shown.
Post-conditions	Client is informed of suitable times to book the caretaker.

Use Case ID	C-07
Use Case Name	Book Services
Primary Actor/s	Client
Description	The client books a selected caretaker for a specific time and date.
Pre-conditions	Client is logged in and has selected a caretaker and time slot.
Main Scenario	- Client confirms service details. - System saves the booking and sends confirmation.
Exceptions	- If a slot is already booked, the client is notified and asked to choose another.
Post-conditions	A new booking record is created and linked to the client and caretaker.

Use Case ID	C-08
Use Case Name	Select Caretaker
Primary Actor/s	Client
Description	The client selects a caretaker from the list for a service.
Pre-conditions	Client is viewing available caretakers.
Main Scenario	- Client clicks on “Select” on a caretaker profile. - Caretaker is linked to the current booking process.
Exceptions	- If the caretaker is no longer available, the system asks the client to pick another.
Post-conditions	The selected caretaker is stored for the booking.

Use Case ID	C-09
Use Case Name	Choose Time
Primary Actor/s	Client
Description	The client selects a preferred time slot for the service.
Pre-conditions	Caretaker's availability has been displayed.
Main Scenario	- Client chooses a date and time from the available options.
Exceptions	- If the time slot becomes unavailable, the client is prompted to reselect.
Post-conditions	Selected time is attached to the booking.

Use Case ID	C-10
Use Case Name	Give Feedback & Rating After Service
Primary Actor/s	Client
Description	The client submits feedback and a star rating after a completed service.
Pre-conditions	The service appointment has been completed.
Main Scenario	- Client navigates to completed booking. - Fills in feedback form and submits.
Exceptions	- If feedback submission fails, the client is prompted to retry.
Post-conditions	Feedback is saved and visible in the caretaker's profile.

Use Case ID	C-11
Use Case Name	View Booking History
Primary Actor/s	Client
Description	The client views a list of all previous and upcoming service bookings
Pre-conditions	The client is logged in.
Main Scenario	- The client navigates to the booking history section. - The system displays past and scheduled bookings.
Exceptions	- If there are no bookings, the system shows a "No records found" message.
Post-conditions	The client can review booking details and service dates.

Use Case ID	C-12
Use Case Name	View Payment History
Primary Actor/s	Client
Description	The client reviews the payment records related to past service bookings.
Pre-conditions	The client has completed at least one paid booking.
Main Scenario	- Client opens the payment history section. - The system shows payment amounts, dates, and statuses.
Exceptions	- If no payment data is available, a message is shown
Post-conditions	The client is informed of their transaction history.

Use Case ID	C-13
Use Case Name	Cancel / Reschedule Contract
Primary Actor/s	Client
Description	The client cancels or reschedules an existing service booking.
Pre-conditions	The client has an active upcoming booking.
Main Scenario	- Client selects a booking to modify. - Client chooses to cancel or reschedule. - System updates the booking status.
Exceptions	- If modification is not allowed (e.g., short notice), an error is shown.
Post-conditions	The booking is either cancelled or updated with a new time.

Use Case ID	C-14
Use Case Name	Contact the Assigned Caretaker
Primary Actor/s	Client
Description	The client contacts the caretaker assigned to their booking.
Pre-conditions	A caretaker has been assigned to the booking.
Main Scenario	- Client opens the assigned caretaker's profile. - Sends a message or request via the platform.
Exceptions	- If the caretaker is not reachable, the system notifies the client.
Post-conditions	The caretaker receives the client's message or request.

Use Case ID	C-15
Use Case Name	Change the Caretakers
Primary Actor/s	Client
Description	The client requests to replace their assigned caretaker with a different one.
Pre-conditions	A current booking has an assigned caretaker
Main Scenario	- Client clicks “Change Caretaker” for a booking. - Selects a new caretaker. - System updates the booking.
Exceptions	- If no alternate caretakers are available, the client is notified.
Post-conditions	The new caretaker is assigned to the booking.

Use Case ID	C-16
Use Case Name	Create Customized Service Plans for Caretaker
Primary Actor/s	Client
Description	The client creates a personalized service plan with selected service types
Pre-conditions	The client is booking a new service.
Main Scenario	- Client chooses service types (e.g., cooking, bathing). - Selects time, days, and preferences. - System saves the custom plan.
Exceptions	- If the chosen options are not supported, the system prompts the client to adjust.
Post-conditions	A custom plan is attached to the service booking.

Use Case ID	C-17
Use Case Name	Filter Caretakers by Location
Primary Actor/s	Client
Description	The client filters the list of available service providers by geographic area
Pre-conditions	The client is logged in and viewing available caretakers
Main Scenario	- Client enters city, district, or postal code. - System filters and displays only relevant profiles.
Exceptions	- If no providers match the filter, a message is displayed.
Post-conditions	The client sees a location-specific list of available service providers.

8.2. Quality attribute requirements

8.2.1. Reliability

The SmartCare system should be highly reliable to ensure continuous service availability for clients, caretakers, and HR managers. It must:

- Function correctly under normal conditions without frequent crashes or data loss.
- Handle failures gracefully, e.g., temporary database issues should not cause the whole system to fail.
- Ensure that caretaker schedules, leave records, and complaints are not lost or corrupted.

8.2.2. Performance

The system must deliver a smooth and responsive experience across all modules. Specifically:

- Pages should load within 2–3 seconds on average.
- Searching caretakers or generating reports must respond quickly, even with moderate amounts of data.

- Simultaneous users (e.g., multiple clients submitting service requests or complaints) should not cause lag.

8.2.3. Scalability

While initially designed for a single caretaker company, the system must be scalable enough to support:

- Increased number of users (e.g., more clients or caretakers in the future).
- Additional service types (e.g., if new services like pet care or nursing are added later).
- Expansion to support more detailed modules such as finance or performance analytics.

8.2.4. Security

As the system handles personal information (user details, schedules, complaints), it must be secure. This includes:

- Role-based access control (Admin, HR, Caretaker, Client).
- Secure login with encrypted passwords and reset options.
- Prevent unauthorized access to sensitive data like salaries and internal reports.
- Sanitize user inputs to protect against SQL injection or XSS attacks.

8.2.5. Maintainability

The system should be easy to maintain and update. This is ensured by:

- Writing clean, modular code using standard web technologies (HTML, CSS, JS, PHP).
- Maintaining good documentation for future developers.
- Using Git for version control to track changes and manage updates.
- Structuring the backend so that service types, roles, and reports can be extended or modified with minimal changes.

8.2.6 Usability

The user interface should be simple, intuitive, and accessible for all user types, including:

- Clients with little technical knowledge (elderly or non-digital-savvy users).
- Caretakers accessing their schedules or submitting leave.
- HR and Admin users needing quick access to reports and management functions.
- Responsive design so it works on desktops, tablets, and mobile browsers.

09. Technologies To Be Used

The **SmartCare – Caretaker Management System** is designed using a combination of front-end and back-end technologies, following the **MVC (Model–View–Controller)** architectural pattern. This structure ensures separation of concerns, scalability, maintainability, and clarity throughout the codebase.

9.1. Front-End Technologies

Used to build the user interface and provide dynamic interactions:

- **HTML5** – For creating structured and semantic markup.
- **CSS3** – For styling the pages and ensuring responsive design.
- **JavaScript** – For enhancing user interaction and front-end validation.

9.2. Back-End Technologies

Used to build the application logic and data processing:

- **PHP (Core PHP)** – Handles server-side logic, authentication, form processing, and role-based access.
- **MySQL** – For storing persistent data such as user accounts, caretaker schedules, feedback, and complaints.
- **Apache HTTP Server** – Acts as the web server to serve PHP pages and static resources.

9.3. Key Components of the Architecture

This system uses a monolithic MVC architecture built with PHP to simplify deployment and development:

1. Authentication

- A custom PHP-based login system that handles secure login, role-based session management, and forgotten password functionality.

2. File Management

- Upload and manage documents such as client feedback files, caretaker profiles, and resumes (if needed for hiring or HR records).

3. Routing

- Basic routing logic to map URLs to specific controllers and views for clarity (e.g., `/caregiver/apply-leave` routes to `LeaveController`).

4. Error Handling and Logging

- System-level error handling and logging to ensure any failures are tracked and recoverable.

9.4. Security Features

Security is a top priority in SmartCare. Key measures include:

- CSRF protection – Prevents Cross-Site Request Forgery attacks in forms.
- Data sanitization and validation – Prevents injection attacks and ensures safe input handling.
- Prepared SQL statements – Protects against SQL injection vulnerabilities by binding parameters properly.

9.5. Performance Optimization

To ensure a responsive experience:

- Basic caching mechanisms will be used for pages like dashboards and reports to reduce server load and improve response time.

9.6. MVC Pattern Overview

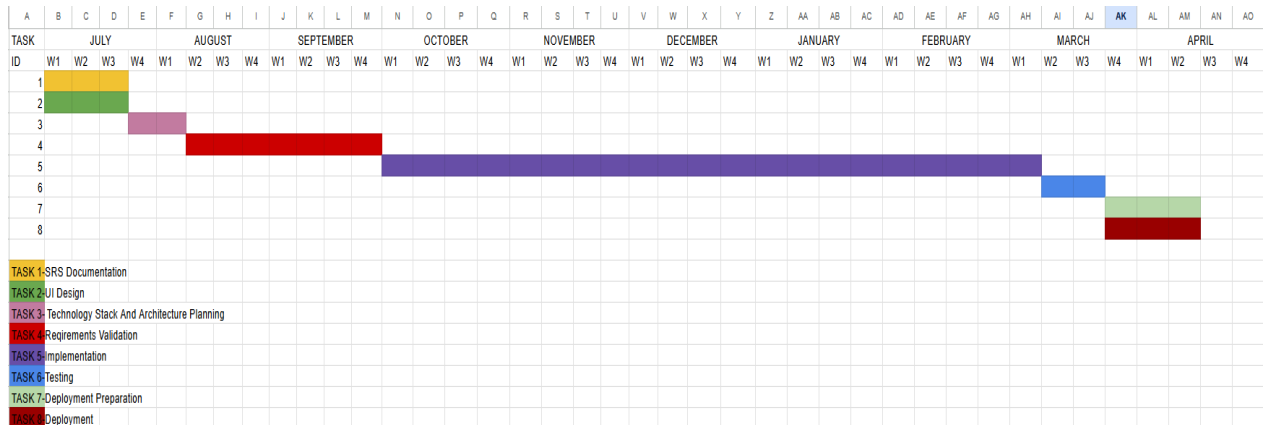
The system is organized using the **Model–View–Controller** pattern:

Layer	Purpose
Model	Interacts with the database (e.g., caretaker data, service requests)
View	Front-end interface for clients, caretakers, HR, and admins
Controller	Handles user actions, communicates with models, and returns views

9.7. Collaboration & Project Management Tools

- Git + GitHub – Version control and collaboration among team members.
- Google Docs – For documentation and report writing.
- Jira / Trello – For task assignment, progress tracking, and sprint planning.
- Figma – For UI design and prototyping.
- Draw.io – For technical diagrams like use cases, flowcharts, and class diagrams.

10. Project Timeline



For the development of our SmartCare – Caretaker Management System, we have chosen to employ the Iterative Development Model. This approach enables us to build the system in incremental cycles, where each iteration delivers a progressively enhanced version of the application.

This model is particularly appropriate for our project due to its dynamic nature and evolving user requirements. It offers several advantages that align well with our objectives:

- **Flexibility:**

The iterative model allows us to adapt to changing requirements and incorporate stakeholder feedback at various stages of the project. This ensures the system remains relevant and meets user expectations.

- **Risk Mitigation:**

By building the system in smaller iterations, potential risks and errors can be identified and addressed early, reducing the impact on overall project quality and timeline.

- **Progressive Functionality:**

Each iteration delivers a working subset of the system — such as login, caretaker scheduling, complaint handling, and feedback modules — which allows for early testing, validation, and feedback from real users.

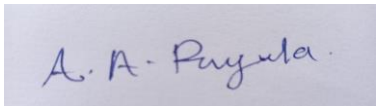
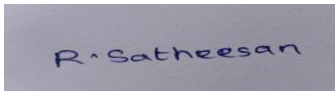
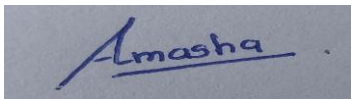
- **Continuous Improvement:**

Based on user feedback and test results from each cycle, we can refine and improve features, fix bugs, and add enhancements — ensuring the system evolves with user needs.

This approach supports our team's development structure and maximizes productivity by focusing on iterative delivery, ongoing feedback, and continuous refinement — making it ideal for the SmartCare system.

11.Declaration

We as members of the project titled **SmartCare – Caretaker Management System**, Certify that we will carry out this project according to the guidelines provided by the coordinators and supervisors of the course as well as we will not incorporate, without acknowledgment, any material previously submitted for a degree or diploma in any university. To the best of our knowledge and brief, the project work will not contain any material previously published or written by another person or ourselves except where due reference is made in the text of appropriate places

Index Number	Name of the student	Signature
23000122	Anne Piyula Arulpirabakar	
23001798	Satheesan Rajasekar	
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